# Curated Research Articles

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- **Synergistic Optimization of Cathode Composite Architecture and Stack Pressure for High-Performance All-Solid-State Chloride-Ion Batteries** — score: 1.000 This study presents a high-performance all-solid-state chloride-ion battery using a VOCl/CsSn . In . Cl3/In architecture, achieving an initial capacity of 169 mAh g−1 and demonstrating durability over 500 cycles. Optimization of the cathode composite composition and stack pressure significantly improved ionic transport and interfacial stability, while advanced techniques revealed key resistance contributions and structural changes during operation, paving the way for more efficient anionic shuttle batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Avalanche-like intercalation and intraparticle correlations in graphite** — score: 1.000 The study utilizes operando optical microscopy to reveal that lithium deintercalation and intercalation in graphite occur through avalanche-like processes, highlighting the influence of local disorder on phase-transition dynamics and ion transport within lithium-ion battery electrodes. This insight could enhance understanding of battery performance and efficiency. Journal: *Nature*
- **SOC Mismatch–Driven Inter-Particle Synchronization via Internal Li-Ion Transfer: An Unrecognized Electrode-Emergent Degradation Pathway in Ni-Rich Composite Cathodes** — score: 1.000 The study reveals that degradation in Ni-rich composite cathodes of lithium-ion batteries is significantly influenced by inter-particle state-of-charge (SOC) mismatches, driven by kinetic variations within the electrode. This internal lithium transfer leads to accelerated surface degradation and challenges the conventional understanding of degradation as merely the result of individual particle failures, emphasizing the need to consider reaction heterogeneity in electrode design. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Orbital-Regulated Stepwise Polysulfide Catalysis Enabled by Ni–Mn D-Electron Complementarity in Aqueous Polysulfide-Based Flow Batteries** — score: 1.000 The study introduces a novel NiS/MnS heterostructure that employs orbital-regulated stepwise catalysis to enhance polysulfide conversion in aqueous redox flow batteries. By leveraging the distinct d-electron configurations of nickel and manganese, the design promotes efficient charge transfer and catalysis, resulting in improved energy efficiency, power density, and cycling stability in battery applications. This approach offers a significant advancement in electrocatalyst design for energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Liquid-Phase Spatial Confinement Engineering of sp2-Rich Closed-Pore Hard Carbon for Sodium-Ion Batteries** — score: 0.900 The article presents a novel solvent-mediated liquid-phase spatial confinement approach to create sp2-rich closed-pore hard carbon anodes for sodium-ion batteries. This method effectively enhances capacity and charge transport while maintaining structural integrity, resulting in high capacities and impressive cycling stability. The engineered carbon demonstrates significant advancements in energy density and operational stability for practical battery applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Stereoelectronic Additive Engineering Enables Solvation Regulation and Dual-Interphase Stabilization for High-Voltage and High-Temperature Lithium Metal Batteries** — score: 0.900 The article presents the use of a stereoelectronically engineered additive, difluorotrimethylsilanylacetic acid ethyl ester (DTAE), to enhance the performance of high-voltage lithium metal batteries (LMBs). By modifying the solvation of Li+ ions and stabilizing the interphases on both cathodes and anodes, the additive significantly mitigates issues like transition-metal dissolution and uneven lithium deposition, leading to improved electrochemical stability and capacity retention in challenging conditions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Anchoring Atomic Ag in Ordered Oxygen Vacancies for High-Performance Anode-Free Lithium Metal Batteries** — score: 0.900 The article presents a novel strategy for enhancing anode-free lithium metal batteries by creating a three-dimensional CeO2 framework with atomically dispersed silver on copper foil. This framework contains ordered oxygen vacancies that improve the lithiophilicity

of the current collector, leading to uniform lithium deposition and significantly reduced dendrite formation. The resulting batteries exhibit high efficiency and cycle stability, demonstrating the effectiveness of interface engineering through atomic rearrangement and metal doping. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Diffusion-like overpotentials from non-diffusion mechanisms in battery particles** — score: 0.900 The article examines how low-frequency electrochemical responses in lithium-ion battery electrodes, typically associated with solid-state lithium diffusion, can instead arise from non-diffusion mechanisms such as electrolyte penetration in porous cathode particles. This insight challenges traditional interpretations of battery performance and highlights the importance of considering alternative factors influencing electrochemical behavior. Journal: *Nature Energy*
- **Closed-loop battery recycling through cathode resynthesis** — score: 0.800 The article discusses advancements in closed-loop battery recycling, focusing on the conversion of black mass into high-quality cathode precursors that meet commercial standards. It examines the relationships between leachate chemistry, impurity levels, and the characteristics of resynthesized materials, while providing design principles aimed at improving hydrometallurgical processes for better cathode performance. Journal: *Nature Reviews Materials*
- **Guiding Successive Mo 4d–S 3p Orbital Hybridization in MoS2 via Dual-Confinement Engineering for Efficient Pseudocapacitive Storage** — score: 0.800 The study introduces a dual-confinement strategy for enhancing the electrochemical performance of layered MoS2 by using lattice-confined manganese (Mn) doping and interlayer-confined ethylene glycol (EG). This innovative approach improves orbital hybridization, increases ionic transport, and results in significant boosts in pseudocapacitive performance with high areal capacitance and remarkable cycling stability for potential supercapacitor applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Enabling Durable, Subzero Li/Na-Metal Batteries with Scalable, In Situ Ultrathin Polymerized Electrolytes** — score: 0.800 The article presents a novel approach to producing ultrathin (approximately 10 µm) solid polymer electrolytes for lithium and sodium metal batteries through a scalable in situ polymerization method. By incorporating potassium additives and fluorine-rich compounds, the electrolytes enhance ion transport, suppress dendrite growth, and reinforce interphases, allowing for durable battery performance under high mass loadings and low temperatures. The developed batteries exhibit impressive cycling stability and conductivity, paving the way for practical applications in high-energy-density energy storage systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Lithium Difluorophosphate and Fluoroethylene Carbonate Containing Electrolytes: How Instrumental Analytics Helps to Understand Synergistic Effects on Performance Improvement in High-Voltage Li Battery Applications** — score: 0.800 This study explores the effectiveness of electrolyte formulations in enhancing the performance of high-voltage lithium-ion batteries by examining the effects of fluoroethylene carbonate (FEC) and lithium difluorophosphate (DFP) additives. The research demonstrates that specific combinations of these components significantly mitigate issues like transition metal deposition and loss of capacity, leading to improved cycling stability and retention through synergistic interactions, as highlighted by instrumental analysis. Journal: *Wiley: Small Methods: Table of Contents*
- **Molecular Engineering of Polytriphenylamine: A Fluorinated Cathode for High-Voltage, Ultrafast, and Stable Sodium-Ion Batteries** — score: 0.800 The article presents a novel molecular engineering approach using fluorinated polytriphenylamine (PFTPA) to enhance the performance of sodium-ion batteries. By modifying the polymer's electronic structure through fluorination, the resulting cathode achieves a high average discharge voltage of 3.75 V, exceptional rate capabilities up to 200 C, and maintains stability over 10,000 cycles, making it a viable candidate for sustainable battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Sustainable Upcycling of Spent LiCoO2 Toward High-Stability NCM Cathode via Eutectic Molten Salt Synthesis** — score: 0.800 The article presents a novel method for upcycling spent

LiCoO2 materials from lithium-ion batteries into high-performance NCM111 cathodes using a LiOH-LiNO3 eutectic molten salt. This approach enables simultaneous lithium replenishment and structural enhancement, resulting in cathodes with superior electrochemical properties, including a specific capacity of 161.6 mAh g−1 and 92.4% capacity retention after 200 cycles, marking a significant advancement in sustainable battery recycling techniques. Journal: *Wiley: Small Methods: Table of Contents*

- **Magnetic Field-Tailored Copper Layers With Fast Dissolution-Redeposition Kinetics for Stable Zinc Metal Anodes** — score: 0.800 This study introduces a magnetic field-assisted approach to enhance the nucleation and growth of copper on zinc anodes, resulting in a uniform protective copper layer. The magnetohydrodynamic effect improves copper ion transport and suppresses detrimental reactions, leading to a robust CuZn5 alloy that significantly enhances the anode's cycling stability and rate capability, achieving over 5000 hours of cycle life at high current densities. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Critical Length Scales to Inhibit Dendritic Fracture in Ion-Exchanged LLZTO** — score: 0.800 The study investigates the effectiveness of ion-exchange (IX) in LLZTO, specifically swapping Li+ for Ag+, to enhance the material's ability to suppress lithium dendrites in batteries. By introducing compressive stress at critical depths through this method, the research achieves a threefold improvement in electrochemical performance while maintaining the structural integrity of LLZTO, offering a promising strategy to enhance the safety and longevity of lithium metal batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Solvation Structure Tailoring of Electrolyte Based on Fluorine-Free, Light-Weight Cyclic Ether Solvent Enables Stable Lithium Metal Anode and Sulfur Cathode Interfaces in Lithium-SPAN Batteries** — score: 0.800 The article discusses how modifying the solvation structure of electrolytes using a lightweight, fluorine-free cyclic ether solvent leads to enhanced stability at the interfaces of lithium metal anodes and sulfur cathodes in lithium-SPAN batteries. This development promises improved performance and longevity for these battery systems. Journal: *ScienceDirect Publication: Nano Energy*
- **Solvent-bridged electrolytes for high-energy Li-ion batteries under extreme conditions** — score: 0.600 The article discusses the development of solvent-bridged electrolytes for lithium-ion batteries, which enhance ionic conductivity and improve performance in extreme conditions. By broadening the liquid range of LiPF6–ether electrolytes, these new formulations mitigate the limitations of traditional LiF-forming electrolytes, enabling better fast-charging and low-temperature operation. Journal: *Nature Chemistry*
- **Polymer Electrolytes for Energy Storage and Electrochromic Energy Devices: From Design Strategy to Multifunctional Applications** — score: 0.600 The article reviews the pivotal role of polymer electrolytes in both energy storage and electrochromic devices, highlighting their essential characteristics that facilitate ion transport and stability while meeting design requirements for optical functionality. It categorizes various types of polymer electrolytes and discusses how their design must adapt to meet the dual demands of electrochemical performance and transparency in applications like smart windows and electrochromic batteries. This framework aims to guide the development of future multifunctional energy systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Electrochemical activity and utilization in iron-air batteries governed by grain size** — score: 0.600 The article explores how the electrochemical performance and efficiency of iron-air batteries are influenced by the grain size of the materials used. It highlights the relationship between microstructural characteristics and battery performance, providing insights that could lead to the optimization of iron-air battery technologies for enhanced energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Efficient battery electrolyte design for electric aircraft driven by physics-augmented AI** — score: 0.600 The article proposes a novel physics-augmented AI framework aimed at efficiently designing battery electrolytes for electric aircraft. This approach addresses the challenges posed by the vast compositional space, reducing development costs while enhancing the efficiency and reliability of

electrolyte design, ultimately contributing to the advancement of eVTOL technology. Journal: *Joule*

- **Engineering periodic Ca2+ doping to topologically reconfigure Na+/vacancy ordering in P2-layered cathode materials for sodium-ion batteries** — score: 0.500 The article discusses the engineering of periodic calcium (Ca2+) doping in P2-layered cathode materials for sodium-ion batteries, aimed at altering Na+ and vacancy ordering. This reconfiguration is expected to enhance the performance and efficiency of these battery materials, with implications for future energy storage technologies. Journal: *ScienceDirect Publication: Nano Energy*
- **Addressing Reproducibility of MXene Synthesis Using Electrochemical and Chemical Treatment of Ti3AlC2 With Tetramethylammonium Hydroxide** — score: 0.400 The article investigates the reproducibility of MXene synthesis from Ti3AlC2 and reveals that rigorous grinding of the precursor powder is essential to break the passivation layer and enhance the etching process with tetramethylammonium hydroxide (TMAOH). It also challenges the effectiveness of electrochemical treatment in NH4Cl/TMAOH, suggesting that successful MXene formation may stem from the chemical etching properties of TMAOH rather than the electrochemical process itself. Journal: *Wiley: Small Methods: Table of Contents*
- **Designing donor-acceptor covalent-organic framework cathode to suppress polyiodide shuttle effect for aqueous zinc-iodine batteries** — score: 0.400 This article discusses the development of a donor-acceptor covalent-organic framework as a cathode material aimed at reducing the polyiodide shuttle effect in aqueous zinc-iodine batteries. The research highlights the material's potential to improve battery efficiency and longevity, addressing a significant challenge in energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Vanadium tailored molybdenum-containing gradient heterostructure for mitigating Jahn-Teller distortion and decoupling competitive ion intercalation** — score: 0.400 This article discusses the development of a gradient heterostructure designed with vanadium and molybdenum to address the challenges posed by Jahn-Teller distortion and improve ion intercalation processes. The research highlights how this tailored material can enhance energy storage capabilities while mitigating structural issues that typically hinder performance. Journal: *ScienceDirect Publication: Nano Energy*
- **Mechanism guided design of Te modified carbon hosts via p–band center modulation for ultra stable lithium–sulfur batteries** — score: 0.400 This article discusses the design of tellurium-modified carbon hosts through the modulation of p-band centers to enhance the stability of lithium-sulfur batteries. The authors propose a mechanism-guided approach that aims to improve battery performance, particularly focusing on increasing the longevity and efficiency of energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Coupled electro-chemo-mechanical degradation in lithium-ion batteries under forced vibration: a multiphysics analysis** — score: 0.300 The article presents a multiphysics analysis of the combined effects of electrochemical and mechanical degradation in lithium-ion batteries subjected to forced vibration. The study explores how these interactions impact battery performance, offering insights for improved energy storage material stability under dynamic conditions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **An open access solid-state battery cell database to advance visualization, search, analysis, and AI data extraction** — score: 0.300 The article introduces a comprehensive open-access database for solid-state battery cells, addressing the challenges posed by the diverse and inconsistent data found in existing literature. It emphasizes the importance of standardizing data formats and practices to enhance accessibility, visualization, and analysis, thereby fostering advancements in battery research aligned with FAIR data principles. Journal: *Joule*
- **Unlocking high power in membraneless Zn–air batteries: A paradigm shift via wireless bipolar electrochemistry** — score: 0.300 The article explores a new approach to enhance the performance of membraneless zinc–air batteries by employing wireless bipolar electrochemistry. This innovative method promises to significantly increase power output, marking a potential breakthrough in energy storage technology. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Tracking thermophysical drift improves latent thermal-runaway risk evaluation in lithium iron phosphate energy storage batteries under low-rate overcharge** — score: 0.300 The article discusses how monitoring thermophysical changes in lithium iron phosphate batteries enhances the assessment of risks associated with latent thermal runaway during low-rate overcharging. The study emphasizes the importance of tracking these changes to improve the safety and reliability of energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Tailoring the Orbital Overlap of RuO2 via Inductive Effect for Durable Proton Exchange Membrane Water Electrolysis** — score: 0.200 This study demonstrates that doping RuO2 with Hf significantly enhances its stability and efficiency as a catalyst in proton exchange membrane water electrolysis. By leveraging Hf's inductive effect to weaken Ru–O covalency, the modified Ru0.7Hf0.3O2 catalyst achieves impressive performance metrics, including 3 A cm−2 at 1.71 V and remarkable durability, operating for over 250 hours with minimal degradation. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Author Correction: A ductile solid electrolyte interphase for solid-state batteries** — score: 0.200 The author correction addresses updates to the findings on a ductile solid electrolyte interphase, which is crucial for improving the performance and longevity of solid-state batteries. This correction clarifies previously reported data, enhancing the understanding of the electrolyte's mechanical properties and their implications for battery technology. Journal: *Nature*
- **Engineering Co–O–Ru Motifs in RuO2 Nanosheets via Asymmetric Spin Polarization for High-Performance and Durable PEM Water Electrolysis** — score: 0.200 The article presents the development of a Co1-RuO2 catalyst, achieved by incorporating isolated cobalt atoms into the RuO2 lattice, which creates Co–O–Ru motifs that enhance asymmetric spin polarization. This innovation significantly improves the catalyst's performance and durability for proton exchange membrane water electrolysis, overcoming the activity-stability trade-off seen in traditional RuO2 catalysts, demonstrating exceptional stability and efficiency in both lab and practical applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Electronic Locking of Labile Cobalt Spinel by Highly Dispersed Iridium for Stable Acidic Water Oxidation** — score: 0.200 The article discusses the development of the Ir-Co3O4 catalyst, which incorporates trace amounts of iridium within cobalt spinel nanosheets to enhance stability in acidic environments during the oxygen evolution reaction (OER). By leveraging an electronic locking strategy, the catalyst achieves high efficiency and remarkable durability in proton exchange membrane water electrolyzers (PEMWEs), demonstrating a significant advancement in utilizing non-noble metal catalysts for sustainable hydrogen production. Journal: *Wiley: Small Methods: Table of Contents*
- **Hydrothermal-Assisted In Situ Intercalation Synthesis of Ti3C2Tx MXene in 6 Hours** — score: 0.200 This article introduces a rapid and scalable method for synthesizing Ti3C2Tx MXene using hydrothermal-assisted etching combined with in situ intercalation, drastically reducing the synthesis time to 6 hours. The approach facilitates effective aluminum removal and delamination under mild conditions, resulting in high-quality, monolayer MXene nanosheets with excellent electrical conductivity, making it a promising route for commercial production. Journal: *Wiley: Small Methods: Table of Contents*
- **Binder Free NiP-Textile Electrodes for Ultrahigh Capacity and 40,000 Cycle Stability in Battery Supercapacitor Hybrid** — score: 0.200 The article reports on the development of binder-free nickel phosphide-coated cotton cloth electrodes, which achieve ultra-high areal capacitance and exceptional cycle stability, making them suitable for flexible energy storage applications. This innovative textile electrode utilizing electroless and electrodeposition techniques shows promising performance in a battery-supercapacitor hybrid device, successfully powering LED panels for extended periods while maintaining efficiency even after extensive mechanical bending cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Toward room temperature fluoride-ion batteries: quaternary ammonium-bearing fluoride-exchange membrane as a quasi-solid polymer electrolyte** — score: 0.200 The

article discusses the development of a quaternary ammonium-bearing fluoride-exchange membrane designed to function as a quasi-solid polymer electrolyte for fluoride-ion batteries, targeting improved performance at room temperature. The research aims to enhance the efficiency and viability of these batteries, contributing to advancements in energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Dual-Functional Ru2P Nanoparticle-Decorated Ru/N,P-Doped Carbon Nanofibers for Efficient Hydrazine Oxidation-Assisted Hydrogen Production and Zinc-Air-Hydrazine Battery** — score: 0.200 The article presents Ru2P nanoparticle-decorated Ru/N,P-doped carbon nanofibers as a highly efficient dual-functional electrocatalyst for hydrazine oxidation and hydrogen production. The synthesized material achieves remarkable performance, requiring minimal potential for hydrazine oxidation and demonstrating superior hydrogen evolution under industrial conditions. Additionally, it enables an innovative water-splitting system while powering a rechargeable Zn-air-hydrazine battery, marking a significant advancement in energy conversion and storage technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **From wasteland to wonderland: Releasing resupply potential from urban minerals of electric vehicle batteries and motors in China** — score: 0.200 The study by Xiong et al. emphasizes the potential of recycling end-of-life electric vehicles in China as a valuable source of urban minerals, particularly lithium, cobalt, neodymium, and dysprosium. However, it also identifies significant deficiencies in the collection and recycling of nickel and manganese, underscoring the necessity for improved policies on recycling and recovery of rare earth elements. Journal: *Joule*
- **A mathematical criterion for chunk susceptibility in Mg–Ca alloy anodes for mg–air batteries: A phase-field modelling and experimental validation** — score: 0.200 The article presents a mathematical criterion to assess chunk susceptibility in Mg–Ca alloy anodes used in Mg–air batteries. It combines phase-field modeling with experimental validation to enhance the understanding of the alloy's behavior under operational conditions, potentially improving battery performance and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Pyridoxal photoenzymes for asymmetric radical–radical cross-couplings** — score: 0.000 The study explores the use of pyridoxal-dependent photoenzymes as catalysts for facilitating asymmetric radical–radical cross-coupling reactions. These reactions hold potential for advancing synthetic methodologies in organic chemistry by providing an efficient and selective means to form complex molecular architectures. Journal: *Nature*
- **Phase-homogeneous mixed halide perovskites for stable tandem photovoltaics** — score: 0.000 The article discusses the development of phase-homogeneous mixed halide perovskites, emphasizing their potential to enhance the stability and efficiency of tandem solar cells. By addressing the challenges of phase segregation in these materials, the research aims to pave the way for more reliable and high-performing photovoltaic technologies. Journal: *Nature*
- **The existential choice facing UK physics facilities: commercialize or close** — score: 0.000 The UK's national synchrotron source, laser facility, and particle accelerator face potential shutdown due to funding shortfalls, prompting a critical decision on whether to pursue commercialization to secure financial support or risk permanent closure. Journal: *Nature*
- **'Timer' in the brain tracks sleep — and predicts awakening** — score: 0.000 Research on mice indicates that chemical modifications on proteins may serve as a potential biomarker for sleep deprivation, enhancing our understanding of how the brain tracks sleep and predicts awakening processes. Journal: *Nature*
- **When should you treat a fever? Evolution holds the answer** — score: 0.000 The article discusses how insights from evolutionary biology can inform better strategies for managing fevers, highlighting the potential for applying natural selection principles to enhance medical care, develop effective social policies, and improve agricultural productivity. Journal: *Nature*
- **How to prevent AI from harming mathematics** — score: 0.000 The article emphasizes the need

for mathematicians to proactively address the potential negative effects of artificial intelligence on the discipline. It argues for immediate action to safeguard mathematical practices and integrity from the challenges posed by AI advancements. Journal: *Nature*

- **Researchers are building computers that run on brain organoids — but have neglected a major ethical issue** — score: 0.000 Researchers are developing biocomputers powered by brain organoids derived from human tissue samples, raising significant ethical concerns about the consent and awareness of individuals providing those samples. The study emphasizes the importance of addressing these ethical issues to ensure transparency and respect for donors in biomedical research. Journal: *Nature*
- **A fever dream of jellies and slugs** — score: 0.000 The article highlights the artistic and scientific contributions of Ernst Haeckel, showcasing his intricate sketches that capture the diverse spectrum of life, ranging from tiny microorganisms to grand organisms. It emphasizes the intersection of art and biology through Haeckel's work, encouraging appreciation for nature's complexity and beauty. Journal: *Nature*
- **From takeout to takeoff** — score: 0.000 Research highlights innovative catalysts utilizing atomically-dispersed ruthenium on oxide supports that efficiently convert plastic waste into jet fuel components under mild conditions, addressing the challenges of high pressure and lengthy reaction times typical in previous methods. This advancement could significantly contribute to decarbonizing the aviation sector. Journal: *Nature Energy*
- **Quasi-ballistic ion transport in a vertical microrod enabling efficient evaporation-driven power generation** — score: 0.000 Wu et al. present an innovative vertical microrod generator that enhances evaporation-driven power generation by optimizing ion transport through methods guided by machine learning. This design addresses the traditional limitations of non-directional fluid flow, resulting in improved efficiency and power density for these energy systems. Journal: *Nature Energy*